

Nuclear energy: Where next?

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As the world accelerates its transition toward a carbon-neutral future, the demand for clean, reliable, and sustainable energy sources is intensifying. Nuclear energy stands out for its high energy density and near-zero carbon emissions. According to the International Energy Agency, it provides about 10% of global electricity and is expected to play a crucial role in achieving net-zero emissions. Yet its long-term sustainability depends not only on operational safety, but also on resource security and environmental responsibility.

URANIUM RESOURCES AND SUPPLY CHALLENGES

Today, nuclear energy predominantly relies on the fission of U-235, which has supported stable electricity generation for decades. Yet land-based uranium resources are both limited and geographically concentrated, with Australia, Kazakhstan, and Canada together producing nearly two-thirds of the world's uranium (World Nuclear Association). These reserves, totaling roughly 8 million tons, are estimated to support global nuclear power generation for only about 70 years (International Atomic Energy Agency and Nuclear Energy Agency). With such limited availability, the planned global expansion of nuclear capacity underscores the urgent need to diversify fuel sources beyond conventional land-based uranium.

DIVERSIFIED RESOURCE STRATEGIES

To address these constraints, researchers and industry stakeholders are actively exploring diversified strategies to enhance the sustainability of nuclear fuel cycles. These efforts aim to expand uranium availability while improving energy security and reducing environmental and geopolitical risks. Key strategies include seawater uranium extraction, thorium-based fuels, and transmutation of long-lived radionuclides, together with improved spent fuel management (Figure 1).

The ocean contains an estimated 4.5 billion tons of uranium—over a thousand times higher than known land-based reserves—making seawater a potentially inexhaustible source for nuclear fuel.¹ As terrestrial reserves are limited and conventional mining poses environmental risks, ocean extraction offers an attractive long-term alternative. Beyond its vast supply, it could reduce geopolitical dependency and support global decarbonization as a cleaner sub-

stitute for mining. However, the extremely low concentration of uranium in seawater (~3.3 µg/L) makes its extraction a grand global challenge, widely recognized for its transformative potential in future energy systems. Extracting uranium at meaningful rates while maintaining economic feasibility and minimizing environmental impacts remains a formidable technical hurdle that demands innovative materials and scalable process engineering.

The thorium molten salt reactor (TMSR) has been proposed as another promising option, as thorium resources are more abundant and widely distributed than uranium, offering opportunities to diversify fuel supply chains and enhance energy security. In a TMSR, Th-232 absorbs a neutron and converts into U-233, a fissile isotope capable of sustaining nuclear fission while potentially reducing the production of long-lived radionuclides. Furthermore, the thorium fuel cycle can reduce the accumulation of transuranic elements in spent fuel, potentially simplifying waste management. Despite these theoretical advantages, thorium-based nuclear technology has not yet achieved commercial deployment. Key challenges include the need for optimized reactor designs to efficiently utilize thorium, the complexity of fuel fabrication and reprocessing, and the proliferation risks associated with U-233. As a fissile material, U-233 can potentially be used in nuclear weapons, raising widespread international concerns over its misuse. Currently, TMSRs remain at the early pilot stage, requiring substantial research and development as well as engineering validation. As such, thorium fuel remains the subject of ongoing research rather than a viable near-term option for nuclear power generation.

Transmutation aims to address the challenge of long-lived radionuclides in spent nuclear fuel by converting them into shorter-lived or stable isotopes, thereby reducing the long-term radiotoxicity and disposal risks. Spent fuel reprocessing, in contrast, enables the recovery of valuable radionuclides such as uranium and plutonium, contributing to a more closed and sustainable nuclear fuel cycle. By recovering resources and reducing waste volumes through reprocessing, and mitigating radiotoxicity through transmutation, these strategies can improve resource utilization and lessen dependence on additional uranium mining. While reprocessing has been industrialized in some countries, transmutation still faces major technical and economic hurdles. Nevertheless, both approaches involve significant technical complexity, high costs, and

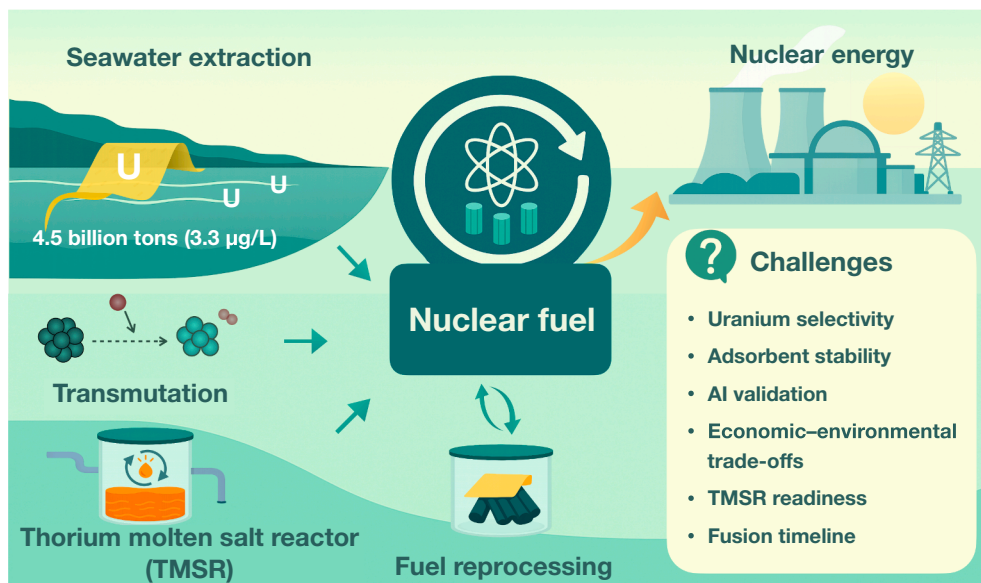


Figure 1. Diversified strategies for sustainable nuclear energy, such as seawater uranium extraction, thorium molten salt reactors, transmutation, and fuel reprocessing, together with key challenges

persistent public concerns over nuclear safety and waste management. Scaling up would require major infrastructure investments, robust regulatory frameworks, and broad societal acceptance.

OPPORTUNITIES AND CHALLENGES IN SEAWATER URANIUM EXTRACTION

From the abovementioned aspects, seawater uranium extraction is attracting renewed attention as a supplemental resource that could strengthen the uranium supply chain. Research efforts increasingly focus on developing materials and systems that can efficiently extract uranium under real marine conditions, addressing both technical feasibility and long-term sustainability.^{2–5}

Porous adsorbents

Amidoxime-functionalized polymers, hydrophilic metal-organic frameworks, covalent-organic frameworks, and bio-based adsorbents have demonstrated promising uranium uptake at the laboratory level. However, competing ions—notably V(V) and Cu(II)—in seawater can interfere with uranium sorption, while biofouling and mechanical wear degrade long-term operational stability. Developing adsorbents with high selectivity, rapid adsorption kinetics, mechanical stability, and resistance to harsh marine conditions is a priority. Additionally, strategies to lower synthesis costs, enhance reusability, and facilitate separation from the ocean are critical for large-scale implementation.

Membrane extraction and photo-/electrochemical methods

Membrane separation and photo-/electrochemical strategies have shown potential for uranium recovery from seawater. However, membrane fouling, poor ion selectivity, limited visible light absorption, and electrochemical side reactions must be carefully addressed through materials innovation and process optimization. Bridging the gap between laboratory success and scalable, cost-effective deployment remains a key challenge for these technologies.

AI-driven materials discovery

Machine learning and high-throughput computational screening accelerate the discovery of adsorbent materials by predicting structure-property relationships. By modeling these relationships and leveraging large datasets, researchers can identify promising candidates before time-consuming experimental synthesis. However, translating computational predictions into practical performance under real seawater conditions demands robust validation to ensure durability and applicability within the broader extraction ecosystem.

Beyond technical barriers, the large-scale feasibility of seawater uranium extraction demands comprehensive evaluation. Factors such as cost competitiveness, energy requirements, environmental impacts, and supply chain integration must be carefully evaluated. In addition, the complexity of seawater chemistry, including fluctuations in temperature, salinity, pH, microbial activity, and organic matter, introduces further challenges that must be addressed through advanced materials design and process optimization. Meeting these challenges will require coordinated efforts among chemists, materials scientists, engineers, oceanographers, computational scientists, biochemists, and policy experts, as well as early stakeholder engagement from fundamental research through pilot-scale demonstration to commercialization. Nevertheless, while seawater uranium extraction holds great promise, it still faces substantial challenges spanning from fundamental research to engineering practice.

OPEN QUESTIONS

Despite recent significant progress in nuclear energy technologies, several fundamental challenges persist as follows.

- (1) How can uranium be extracted with high selectivity from real seawater, given its ultralow concentration and the abundance of competing ions?
- (2) Which adsorbent materials and extraction methods are most promising for real application, and how can they retain long-term performance under seawater conditions while overcoming corrosion, biofouling, and scale-up limitations?
- (3) How can AI and machine learning be effectively integrated with laboratory tests and ocean conditions to accelerate the precise design and deployment of materials?
- (4) What are the trade-offs between economic feasibility and environmental impacts in scaling up seawater uranium extraction technologies?
- (5) When can TMSRs become a viable option for nuclear power, and what are the main obstacles still ahead?
- (6) How long will it take before fusion reactors can reliably supply electricity, and how should we plan if this timeline remains uncertain?

CONCLUSION

The future of nuclear energy will rely on a diversified strategy—including seawater uranium extraction, thorium reactors, transmutation, and AI-assisted material discovery—to ensure a secure and sustainable energy supply. While each approach holds promise, all face significant technical and socio-environmental barriers to large-scale deployment. Realizing this goal requires sustained innovation and collaboration. As the world accelerates toward carbon neutrality, the question “nuclear energy: where next?” becomes a pressing call to action. With continued advances, emerging technologies may evolve from concept to practical solutions contributing to a cleaner energy future.

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DECLARATION OF INTERESTS

The authors declare no competing interests.

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